SPE-227287-MS

Multi-Stage Cementing With Metal-to-Metal Supported Packers for Effective CCA Prevention

Hussain Alzain, Luis Urdaneta, and Hany Gamal, Weatherford Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Casing-to-casing annular gas migration is among the most severe problems at gas wells and may occur during drilling or well completion. The possible consequences are diverse and not always immediately detectable. However, severe surface complications, such as gas flow or pressure at the wellhead, may necessitate well abandonment. Thus, elaborate measures are required to secure wells integrity. The current abstract meets this need by elucidating real-life solutions to prevent casing-to-casing annulus pressure in gas wells.

The abstract intends to describe multi-stage cementing through metal-to-metal supported packer (packoff stage tool (POST) also known as differential valve (DV Tool) to prevent casing-to-casing annulus pressure in gas wells. A case study methodology was selected to enable a detailed assessment of the said phenomenon within the Middle East field applications. The proposed technical approach was divided into the foundation, pre-field, field, and reporting phases. The field deployment showed outstanding DV Tools performance in preventing casing-casing annulus pressure while performing multi-stage cementing. Consequently, data and information were collected in the field phase and compiled into meaningful insights within reporting stage.

The case study identified the tool's successful development research and application for the drilling equipment and services. The obtained results indicated the feasibility of successfully isolating multiple zones in gas in 9 5/8-in and 13-3/8-in casing sections, followed by multi-stage cementing (ideally two-stage) to separate lost circulation zones. In this case, correctly placing cement up to the surface level is crucial to provide an extra annular barrier to prevent casing-to-casing annular gas migration. The innovative approach presents the state-of-the-art in preventing casing-to-casing annular pressure in gas wells in Gulf Cooperation Council countries. This research is among the first attempts to summarize the success of using multi-stage cementing with a metal-to-metal supported packer to address the said problem. Thus, the provided information should establish an essential reference for future studies and facilitate the global adoption of similar solutions within the oil and gas fields.

The results of field trials and case studies further demonstrate the effectiveness and economic viability of this approach, validating its potential to transform CCA prevention strategies and elevate wellbore performance to unprecedented levels.

Key Words: Innovative Approach, Multi-Stage Cementing, Metal-to-Metal, Packers, CCA Prevention

Introduction

The integrity of an oil and gas well is a critical cornerstone for safe, efficient, and environmentally responsible hydrocarbon production, a principle underscored by the complex and technically demanding process of well cementing. Cementing, in its essence, is a meticulously engineered operation involving the pumping of a specialized cement slurry into the annular space between the wellbore and the casing, or between concentric casing strings, to achieve several paramount objectives: primary zonal isolation, structural support for the casing, and protection of metallic components from corrosive downhole fluids. Without the rigorous application of effective cementing techniques, wells are highly susceptible to interzonal communication, unwanted fluid migration, and structural instabilities, which collectively lead to significant operational challenges, elevated safety risks, and pronounced environmental concerns. This comprehensive technical guide endeavors to provide an in-depth understanding of the engineering principles, operational intricacies, and new technological advancements specifically targeted at addressing a pervasive and critical challenge in well integrity: casing-to-casing annular (CCA) gas migration, a phenomenon that, if unmitigated, can compromise well safety and environmental compliance.

Well Cementing Engineering in Oil and Gas Operations

Well cementing serves multiple vital functions throughout the entire lifecycle of a well, commencing from the initial drilling phases through to eventual abandonment, with its fundamental purpose revolving around the establishment of a robust hydraulic seal that effectively prevents the uncontrolled movement of fluids, whether hydrocarbons or water, between distinct geological formations or to the surface. This critical function of zonal isolation is not merely an operational nicety but an absolute prerequisite for both the efficient production of hydrocarbons from targeted reservoirs and, equally importantly, the safeguarding of non-target zones, such as sensitive aquifers, from contamination by drilling fluids or hydrocarbons. Beyond its isolating role, cement imparts crucial structural support to the casing, enabling it to withstand the formidable external pressures and forces exerted by the surrounding geology, and concurrently acts as a protective barrier, shielding the casing from corrosive downhole fluids, thereby significantly extending the operational lifespan and integrity of the well.

Cementing operations are broadly bifurcated into primary cementing and secondary (remedial) cementing. Primary cementing is performed immediately following the running of a casing string into the wellbore, representing arguably the most pivotal cementing operation, as its success or failure largely predetermines the long-term integrity, productivity, and environmental footprint of the well. Secondary cementing, conversely, involves remedial actions undertaken to address issues that manifest subsequent to primary cementing, such as deficiencies in zonal isolation, casing leaks, or instances of lost circulation, typically involving techniques like squeeze cementing, wherein cement is forcibly injected into compromised zones or leaks, or plug cementing, utilized for sidetracking or well abandonment. The meticulous engineering design and flawless execution of primary cementing are thus unequivocally essential to preempt future complications, particularly those directly related to gas migration, forming the bedrock of sustainable well operations (Agarwal et al., 2024).

Engineering Cementing Excellence: Workflow-Driven Planning, Execution, and Evaluation for Well Integrity

Figure 1 highlights the systematic approach to achieving flawless primary cementing from meticulous planning and precise execution to rigorous evaluation ensuring long-term well integrity and zonal isolation. The ultimate success of a primary cementing job is inextricably linked to rigorous pre-job planning, precise slurry design, and an unwavering commitment to meticulous execution, a process that commences long before the first volume of cement is mixed, encompassing comprehensive geological and reservoir data analysis, precise wellbore trajectory planning, and a nuanced understanding of anticipated downhole conditions including temperature, pressure, and the characteristics of formation fluids (Upadhyay et al., 2025).

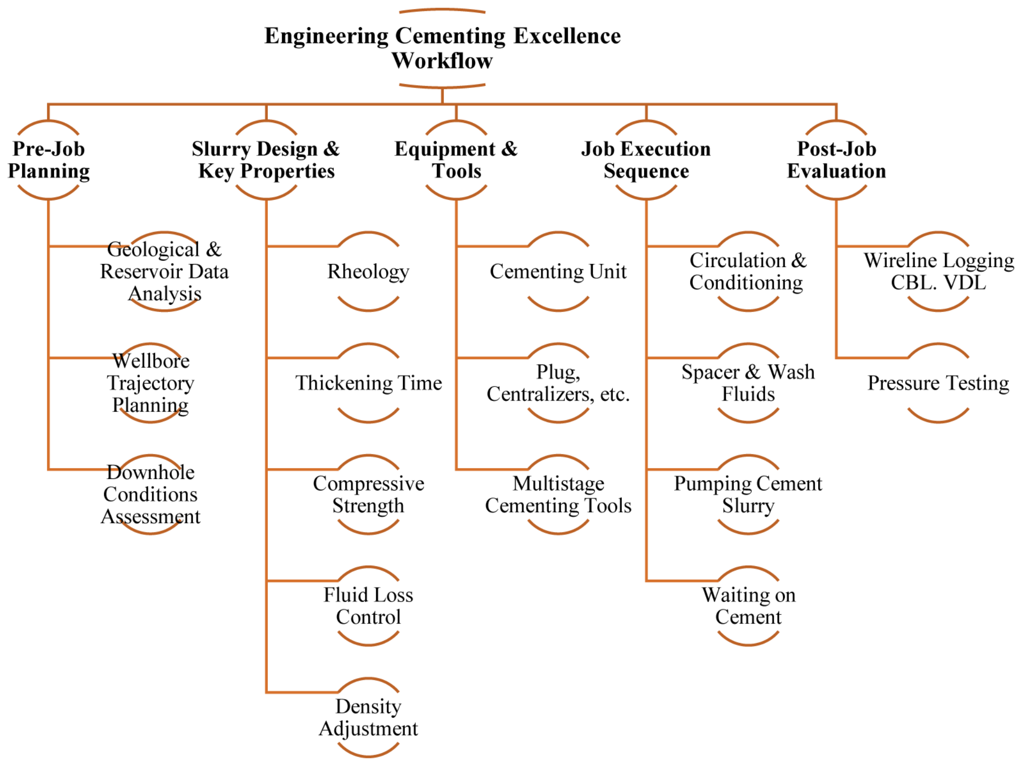

Figure 1—Engineering Cementing Excellence Workflow

A critical aspect of this planning phase is the slurry design, which necessitates the judicious selection of appropriate cement blends and a suite of specialized additives tailored to meet specific downhole conditions and stringent operational requirements. Key properties that must be rigorously considered and optimized include rheology, which governs the flow characteristics of the cement and its efficiency in displacing drilling fluids, demanding a low viscosity during pumping for effective displacement yet requiring the rapid development of sufficient gel strength post-placement to prevent fluid migration. Thickening time defines the duration during which the cement remains pumpable, a parameter that must be meticulously optimized to ensure adequate time for mixing, pumping, and placement without the detrimental effects of premature setting. Compressive strength signifies the set cement's capacity to endure external forces without structural failure, with adequate strength being fundamental for long-term casing support and overall well integrity. Fluid loss quantifies the tendency for water to filter out of the cement slurry into permeable formations, where excessive fluid loss can lead to premature dehydration, compromised bonding, and increased slurry viscosity, thereby necessitating the inclusion of fluid loss control agents to mitigate these risks. Density is a critically important parameter, essential for maintaining hydrostatic control and preventing undesirable formation breakdown or influx, with density adjustments typically achieved through the incorporation of lightweight additives such as microspheres or bentonite, or heavyweight additives like hematite or barite. Finally, bonding properties encapsulate the cement's ability to create a strong, lasting adhesion to both the casing surface and the formation, a characteristic significantly influenced by the precision of slurry design, the efficiency of drilling mud removal, and proper casing centralization. To ensure a uniform and complete cement sheath around the casing, centralizers are strategically installed on the casing string; these mechanical devices are instrumental in maintaining concentric casing positioning within the wellbore, thereby facilitating efficient mud displacement and preventing the formation of detrimental bypass channels, often supplemented by standoff devices to maintain a consistent annulus. The cementing equipment array primarily comprises a cementing unit, typically truck-mounted or skid-mounted, housing high-pressure pumps, mixing tubs, and a densitometer for precise slurry density control, alongside plug containers utilized for launching bottom and top cementing plugs, and displacement tanks for holding the fluid that propels the cement slurry down the casing.

The job execution sequence initiates with circulation and conditioning, wherein drilling fluid is circulated through the annulus to cleanse the wellbore, remove cuttings, and condition the drilling mud for efficient displacement, often involving multiple bottom-up circulations. Spacer and wash fluids are then strategically pumped ahead of the cement slurry, with spacers acting as physical separators to prevent drilling mud contamination of the cement and promote superior bonding, while washes (e.g., water, acid solutions) are employed to meticulously clean the casing and formation surfaces. The core of the operation involves pumping cement, where the cement slurry is mixed on the fly to exacting specifications and pumped down the casing, with a bottom plug launched ahead of the cement to prevent contamination by the displacement fluid. Subsequently, displacement occurs after the designed volume of cement has been pumped, the top plug is launched, and displacement fluid is pumped behind it, propelling the cement slurry up the annulus until the top plug "bumps" against the float collar at the bottom of the casing, signaling the completion of cement placement in the annulus.

A critical period of waiting on cement (WOC) time then ensues, during which the cement is allowed to set and develop sufficient compressive strength, a duration absolutely vital for the cement to attain its full mechanical properties prior to the resumption of drilling activities. Post-job evaluation of cementing success is routinely conducted using wireline logging tools, predominantly the cement bond log (CBL) and variable density log (VDL), which measure the quality of the cement bond to both the casing and the formation, thereby identifying any areas of poor bonding, gas channeling, or voids, with final verification through pressure testing of the casing and annular spaces to confirm zonal isolation and overall well integrity.

Multi-Stage Cementing: An Engineered Solution for Annular Gas Migration Mitigation

Annular gas migration (AGM) remains a significant challenge in gas wells, despite advancements in cementing technology as shown in Figure 2. AGM signifies uncontrolled gas flow in the annular space after primary cementing, manifesting as sustained casing pressure (SCP) or surface casing vent flow (SCVF), posing immediate safety and environmental threats. Key causes include gas channeling due to incomplete mud displacement or insufficient slurry density, and mud contamination compromising cement strength. Poor cement bond and micro-annuli, formed by inadequate adhesion, casing movement, or cement shrinkage, create gas pathways. Insufficient hydrostatic pressure during cement setting, often from fluid loss or gas cutting, allows formation gas invasion. Additionally, overly rapid early gel strength development can prematurely "bridge off" cement, leaving unsealed channels. The consequences are profound, encompassing SCVF (direct gas leakage causing environmental harm via methane emissions and safety hazards like explosions), and SCP, indicating zonal isolation failure. Environmental concerns extend to aquifer contamination, while safety risks include toxic gas exposure. Economic losses from costly remedial workovers and regulatory fines further underscore the criticality of preventing AGM. Detection involves wellhead surveys, gas analysis, and pressure tests to characterize migration.

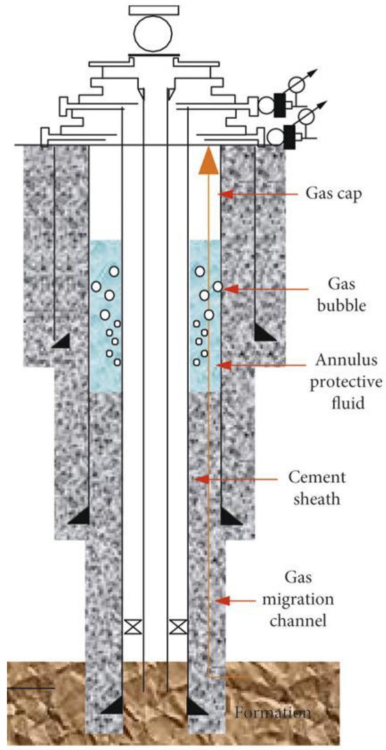

Figure 2—Annular Gas Migration (Feng et al., 2019)

Multi-Stage Cementing DV Tool

To effectively mitigate the pervasive risks associated with AGM, particularly in challenging well environments characterized by extended cementing intervals, naturally fractured formations, or high-pressure gas zones, multi-stage cementing has emerged as a robust and indispensable engineering solution. In stark contrast to conventional single-stage cementing, where the entirety of the cement column is placed in one continuous operation, multi-stage cementing meticulously involves the placement of cement in two or more discrete stages. This sophisticated technique proves exceptionally effective in addressing critical challenges related to excessive hydrostatic pressure exerted on weak formations, mitigating lost circulation events, and, crucially, preventing annular gas migration, thereby reinforcing well integrity (Hussain et al., 2023).

The fundamental concept underpinning multi-stage cementing is the strategic division of a long annular section into shorter, more manageable segments. Cement is initially placed in the lower stage, and after it has achieved sufficient setting (or partial setting), a specialized packoff stage tool [DV tool] is activated to permit the cementing of the subsequent upper stage. This phased approach confers several significant engineering advantages: it fundamentally reduces hydrostatic pressure by segmenting the cement column, thereby minimizing the hydrostatic pressure exerted by the wet cement column on lower, potentially weaker formations, which in turn reduces the risk of formation breakdown and lost circulation, known precursors to gas influx. It facilitates improved zonal isolation by allowing for the tailored design of cement slurries optimized for distinct geological zones; for instance, a dense, high-strength cement can be deployed across the reservoir, while a lighter, more flexible cement can be utilized across shallower, sensitive zones, thereby enhancing the likelihood of achieving complete and durable zonal isolation. Critically, it provides substantial prevention of gas migration by creating multiple independent isolation barriers, significantly reducing the available pathways for gas to migrate up the annulus, as the setting of the first stage cement establishes a vital primary seal before the upper stage is placed. Furthermore, it promotes enhanced wellbore cleaning due to shorter cementing intervals, which facilitate more effective displacement of drilling fluids and superior conditioning of the wellbore, leading to a much improved cement bond. Finally, the technique offers increased flexibility in navigating challenging downhole conditions, as individual stages can be meticulously engineered to specifically address unique problems encountered within different sections of the annulus (Urdaneta et al., 2020).

Figure 3 illustrates a three-stage cementing job through a series of schematics. Panel (a) depicts the initial state where the casing is being run into the wellbore. Panel (b) shows the execution of the first-stage cementing operation, utilizing a bypass plug and a displacement opening plug. Moving to panel (c), the open-hole packer is being set, followed by the opening of the differential valve (DV) tool and the commencement of the second-stage cement job. Panel (d) details the closing of the DV tool and the dropping of a free-fall plug. Panel (e) illustrates the setting of the cased-hole packer and the subsequent opening of the cased-hole DV tool. Finally, panel (f) shows the pumping of the third-stage cement job and the closing of the cased-hole DV tool.

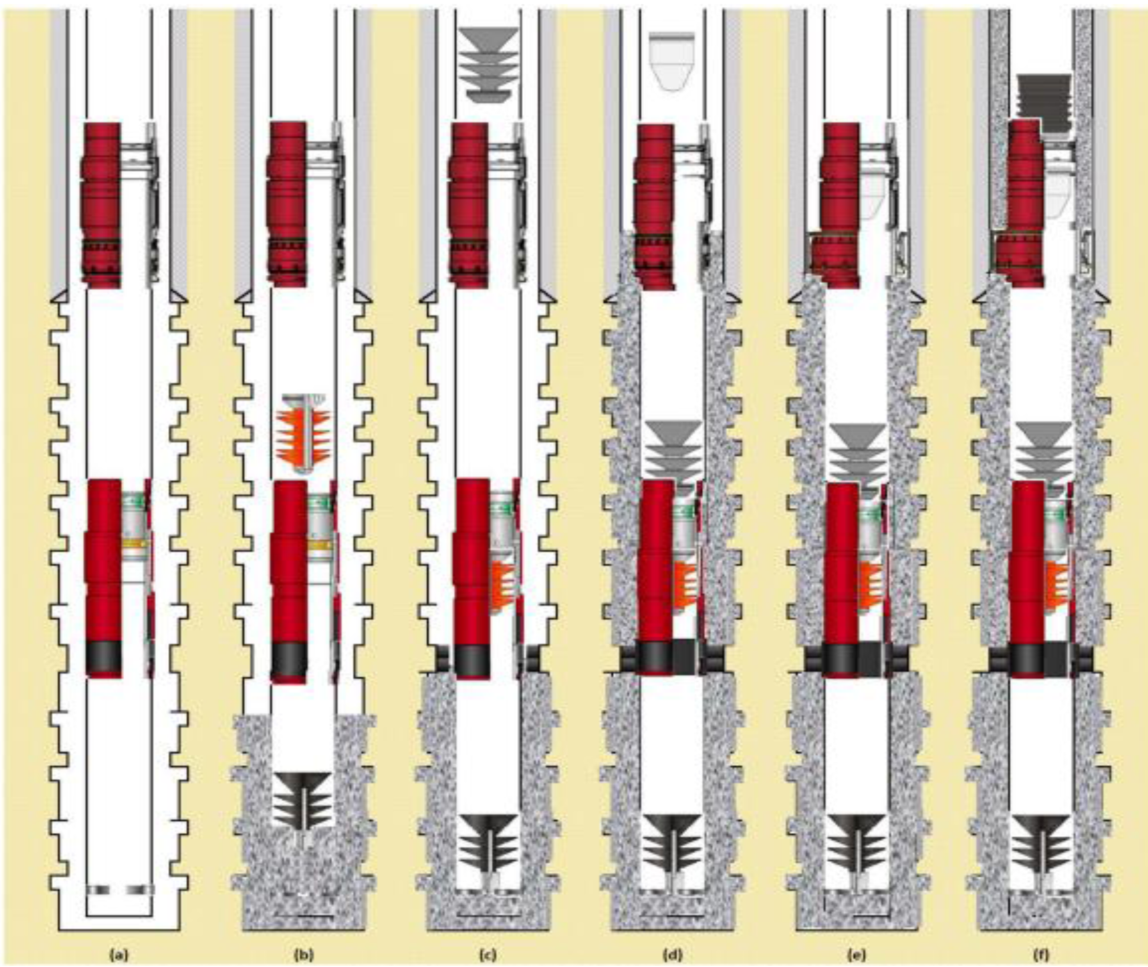

Figure 3—Multi-Stage Cementing (Quintero et al., 2024)

The successful implementation of multi-stage cementing is intrinsically reliant on specialized downhole tools, primarily the multi-stage cementing collar or stage collar, which is strategically incorporated into the casing string at a predetermined depth, typically positioned above the critical zones where the first stage cement will terminate. This tool comprises an outer sleeve and an inner sleeve featuring ports that are initially sealed. The operational sequence typically involves first stage cementing, where cement for the lower section is pumped down the casing and displaced into the annulus below the stage collar, akin to a conventional cementing operation. Subsequently, activating the stage collar occurs after the first stage cement is placed, by dropping a specialized "opening plug" or "dart" from the surface, which lands in the stage collar and, upon applied pressure, shifts the inner sleeve to expose the cementing ports. Once these ports are open, second stage cementing commences, with cement for the upper stage pumped down the casing and exiting through the newly opened ports into the annulus above the stage collar, sometimes followed by a closing plug. Finally, closing the stage collar is achieved by actuating a "closing plug" or similar mechanism to shift the inner sleeve back, sealing the ports and preventing contamination of the set cement, thereby clearing the casing bore for subsequent drilling operations.

Multi-stage cementing is particularly indispensable in deep and ultra-deep wells where a single long cement column's hydrostatic pressure would exceed the fracture gradient of shallow formations, in wells with significant pressure differentials between formations, making single-stage isolation problematic, in gas wells and High-Pressure, High-Temperature (HPHT) wells due to their heightened risk of gas migration, in wells with severe lost circulation zones where multi-stage cementing can isolate these zones, and in environmentally sensitive areas where even minimal gas leaks are intolerable.

Packoff Stage Tool Engineering Design

Unlike standard multi-stage cementing collars that primarily facilitate sequential cement placement, a POST or DV tool integrates a robust mechanical packer assembly. This assembly meticulously creates an exceptionally reliable seal within the annulus, both during and critically after cementing operations (Figure 4). The typical design includes opening and closing sleeves/ports, akin to conventional stage collars, which regulate cement flow for the upper stage. However, the paramount distinguishing feature is the integrated packer element. This element, often a high-expansion packer constructed from elastomer reinforced with metal components or a purely metal-to-metal seal in advanced designs, is engineered to be precisely set and energized against the outer casing or open hole, establishing a formidable physical barrier. The packer's activation involves precise pressure manipulation from the surface, often coordinated with specialized plugs or darts, causing it to radially expand and create an intensely tight seal (Urdaneta et al., 2020).

The unique sealing mechanism inherent to POST/DV tools directly and effectively addresses the chronic problem of casing-to-casing annulus pressure in gas wells. Once set, the integrated packer forms a highly durable and high-integrity mechanical barrier within the annulus, which physically impedes the upward migration of gas or fluids, even in scenarios where the cement sheath itself develops minor imperfections such as micro-annuli or subtle channeling. This definitive seal provides an enhanced layer of zonal isolation at a strategically chosen point, proving particularly effective when precisely positioned at or near the top of a gas-bearing zone or where potential pressure communication pathways are anticipated. By creating such a robust seal, the POST/DV tool significantly aids in pressure containment, effectively confining any residual gas pressure that might develop below the packer, thereby preventing its further migration up the annulus to the wellhead, which is instrumental in maintaining zero sustained casing pressure. Furthermore, the packer provides crucial support for the cement column above it, particularly vital in deviated or horizontal wellbores, ensuring superior cement placement and mitigating the risks of fallback or slumping. Finally, the metal-to-metal design, in particular, offers unparalleled durable sealing in demanding environments, exhibiting superior resistance to the extreme high temperatures, high pressures, and corrosive environments (e.g., H_2S, CO_2) commonly encountered in gas wells, conditions that can rapidly degrade conventional elastomer packers over time, thus ensuring the reliability of the seal for the long-term integrity of the well.

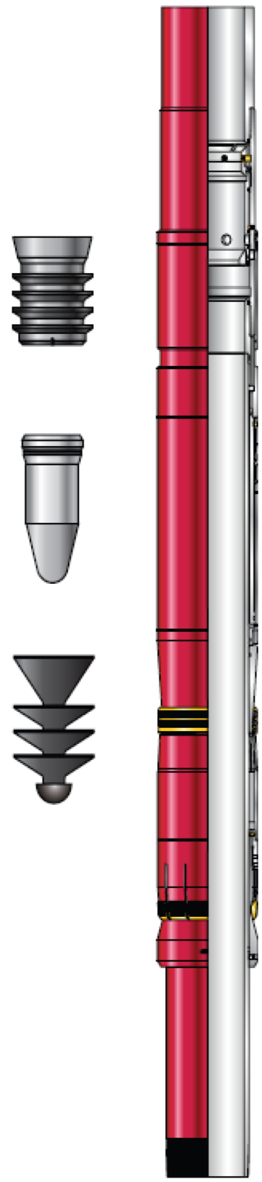

Figure 4—Packoff Stage Tool

Case Study: Comprehensive Technical Approach

The successful implementation of sophisticated cementing solutions, particularly those involving multi-stage cementing integrated with advanced POST/DV tools, absolutely necessitates a highly structured and phased technical approach to ensure project success and long-term well integrity. Figure 5 shows systematic methodological framework is logically segmented into four distinct yet interconnected phases: foundation, prefield, field, and reporting.

Figure 5—Technical Workflow and Engineering Methodology

Phase 1: Foundation (Strategic Planning & Understanding)

This initial phase focuses on comprehensive strategic planning and establishing a fundamental project understanding. It begins with in-depth research and feasibility studies to analyze specific well characteristics, geological formations, reservoir fluid properties, and historical cementing challenges. A rigorous evaluation of the technical and economic viability of multi-stage cementing with POST/DV tools against conventional methods is conducted. This leads to technology selection and evaluation, where appropriate cementing technologies, including specific cement formulations, additives, and specialized downhole tools, are identified through engagement with technology providers and critical review of their specifications and track records. A comprehensive risk assessment and mitigation planning identifies potential operational, environmental, and safety risks (e.g., lost circulation, tool failure, gas influx) associated with the proposed plan, followed by the development of detailed mitigation strategies. Ensuring regulatory review and compliance is paramount, guaranteeing adherence to all relevant requirements and obtaining necessary permits. Finally, precise resource allocation defines the human, financial, and material resources required, including specialized personnel, budget, and logistical considerations.

Phase 2: Pre-Field (Detailed Engineering & Preparation)

This phase translates foundational knowledge into detailed engineering, procurement, and extensive preparation before any on-site operations. It entails developing a detailed engineering design for the comprehensive cementing program. This includes precise wellbore geometry, meticulous cement slurry design (specifying lead and tail slurries, density, rheology, thickening time, compressive strength, fluid loss, and specialized additives tailored for each stage), detailed spacer and wash programs, exact tool placement and operational procedures for POST/DV tools/stage collars, and precise pumping schedules. Optimization of centralizer and standoff placement is also critical. Equipment procurement and preparation ensures all necessary cementing equipment, specialized downhole tools, and accessories are inspected, certified, and ready for deployment. Personnel training and competency assessment provide specialized training on multi-stage cementing and POST/DV tool operation, ensuring readiness. Rigorous logistics planning optimizes transportation and management of all materials. Comprehensive contingency planning develops detailed responses for potential non-routine events. Finally, laboratory testing on cement samples under simulated downhole conditions verifies slurry properties and confirms performance.

Phase 3: Field (Execution & Real-time Management)

This phase represents the actual execution of the cementing operation at the well site. It commences with a thorough pre-job safety meeting to review the program and discuss hazards. This is followed by equipment rig-up and testing to ensure all systems are functional. Wellbore conditioning and mud displacement are executed as designed. The core operation begins with cementing execution (Stage 1), pumping the first-stage cement slurry with close real-time data monitoring. Crucially, the POST/DV tool activation procedure is executed, involving precise pressure control to open the tool and initiate the internal packer setting. Subsequently, cementing execution (Stage 2) involves pumping the upper-stage cement. The final operational step is tool closing to secure the packer and clear the casing. Throughout, real-time monitoring and troubleshooting are paramount, with continuous oversight of all cementing parameters and readiness to implement contingency measures. Quality control involves collecting cement samples for post-job lab testing. The phase concludes with the waiting on cement and subsequent pressure testing of casing and annular spaces to confirm zonal isolation.

Phase 4: Reporting (Evaluation & Lessons Learned)

The final phase focuses on evaluating operational success and documenting invaluable lessons learned. This involves a detailed post-job analysis of all real-time data, post-job logs, and laboratory test results to evaluate cementing performance against design objectives, specifically assessing the POST/DV tool's effectiveness in preventing AGM. Performance evaluation quantifies zonal isolation achievement. Crucially, lessons learned, and best practices are meticulously documented, identifying successes, challenges, and deviations to inform and improve future cementing operations. Comprehensive documentation compiles a final report detailing the entire operation, serving as a vital record for well integrity management. Recommendations for future operations are formulated based on these lessons, guiding optimization. Finally, regulatory reporting ensures all required documentation is submitted, demonstrating adherence to well integrity standards.

This structured, phased approach provides an indispensable framework for managing the inherent complexities of advanced cementing operations, significantly enhancing the probability of achieving long-term well integrity and effectively mitigating critical risks associated with annular gas migration.

Technology Case Study: Tool Application Within Onshore Production Well

Building upon the success of previous workflow and installations, case study for one of the operators designated to deploy an inflatable pack-off stage tool within an onshore production well, targeting challenging drilling and zonal isolation objectives. The primary goals for this application included drilling openhole to total depth, setting the casing string at TD to case off water-sensitive shale intervals, and isolating problematic lost-circulation zones by cementing across critical sections of the formation. The recommended solution incorporated an inflatable pack-off stage tool, strategically set inside the previous casing string. This deployment was enhanced by the addition of an integral torque shoulder and the use of an epoxy compound to provide improved pup joint torque capability, thereby supporting complex downhole operations. The operation commenced with the reaming of the casing string over a 316-foot interval, which was completed within 3.75 hours at a rotational speed of 20 rpm. During this phase, torque measurements varied from 2,000 to 6,000 ft-lb, and weight on bit (WOB) values reached as high as 10,000 lbf. Following this, the casing string was advanced by drilling an additional 301 feet over 14.75 hours, utilizing a higher rotational speed of 50 rpm.

Subsequent to the drilling activities, the team conducted two bottoms-up circulations to ensure wellbore cleanliness before proceeding with the first stage of cementing. At this point, no flow returns were observed at the surface. The pack-off stage process was then initiated by dropping the opening cone, inflating the pack-off tool, and filling the annulus with 140 barrels of mud, after which returns were observed at the surface. To confirm isolation, the team maintained static fluid levels in the annulus for five minutes before opening the pack-off tool. Returns at the surface during this second bottoms-up circulation indicated the pack-off tool had set effectively and zonal isolation was achieved as planned.

Through the successful deployment of the inflatable pack-off stage tool, the engineering team was able to avoid the time-consuming process of pulling the casing and reconditioning the borehole that resulted in an estimated savings of 48 rig hours. Additionally, the operation eliminated the need for a remedial cement job, further optimizing project efficiency and cost. This case demonstrates the technical effectiveness and operational value of incorporating inflatable pack-off stage tools in complex production well environments, particularly where lost-circulation zones and sensitive shale intervals present significant challenges.

Conclusions

This study comprehensively examined the engineering principles and operational challenges associated with well cementing in the oil and gas industry, with a particular focus on multi-stage cementing for effective CCA gas migration.

- Annular gas migration (AGM) remains a significant challenge, especially in gas wells, leading to sustained casing pressure or surface casing vent flow and posing safety, environmental, and economic risks. AGM's causes are complex, including gas channeling, mud contamination, poor cement bonding, micro-annuli, insufficient hydrostatic pressure, and premature cement gel strength development.

- Multi-stage cementing offers a robust solution by allowing tailored cement placement and reducing hydrostatic pressure. Advanced tools like packoff stage tools (POST) / differential valves (DV Tools) significantly enhance zonal isolation through durable metal-to-metal seals, preventing gas migration and containing annular pressure in harsh environments.
- POST/DV tools provide superior zonal isolation, reduce SCP/SCVF, improve safety and environmental compliance, and minimize costly remedial work. Successful implementation relies on a structured engineering workflow.
- Adopting these principles and technologies improves well integrity, reduces environmental impact, and ensures safer, more responsible hydrocarbon production.
- Continuous technological advancements in cement formulations, digital monitoring, and tools engineering designs further enhance well integrity.

Acknowledgment

The authors express sincere appreciation to Weatherford Company for its invaluable support in conducting this study and for granting publication permission.

References

1. Agarwal, R., Essmat, N.Z., Paz, D., Alismail, A., Lopez, G., Diaz, L., Levario, J.A., Corona, M., 2024. Cemented Multi-Stage Fracturing Systems Enhancing Risk Mitigation and Time Efficiency – A Game-Changing Alternative to Open Hole MSF and Cemented Liners.
2. Feng, F., Cong, Z., Shan, W., Hu, C., Yan, M., Han, X., 2019. A Prediction Model for Sustained Casing Pressure under the Effect of Gas Migration Variety. *Mathematical Problems in Engineering* 2019, 3097259. https://doi.org/10.1155/2019/3097259
3. Hussain, M.Q., ALRashdi, A.A., ELMahdy, M.M., Mwansa, P.L., Amorocho, A., Ibrahim, A.M., Naga, H., Kaddoura, I., 2023. Positive Impact of V0 Multi-Stage Cementing Tool on the Long Term Well Integrity.
4. Quintero, E., Martanto, R., Edgar, D., Masoud, R., Urdaneta, L., Martinez, C., 2024. Enhancing Zonal Isolation in Total Loss Zones: Advanced Cementing Techniques for Deep Gas Wells.
5. Upadhyay, N., Al Shehhi, H., Mandal, V., Varma, E., Abdelghany, M.S., Alsaadi, A.M., Maheza, W., Alneyadi, A., 2025. Operational Excellence by Cementing the Hcd-Liner Less 9-5/8Ȳ Casing Design with Single Stage Cementation, Eliminating the Need for Multi-Stage Tool and Dedicated Clean Out Trip in Uae Onshore.
6. Urdaneta, L.A., Farley, D.B., Alzain, H.A., 2020. Case Studies in the Kingdom of Saudi Arabia Demonstrate the Ability to Drill and Ream–in Casing with Multi Stage Cementing Devices and Well Bore Barriers.